

```
import pandas as pd

data = [
    [1, "Romilda", 700.00, 2000.00, 4000.00],
    [2, "Cremilda", 512.00, 3000.00, 4000.00],
    [3, "Lucinda", 784.00, 2500.00, 3000.00],
    [4, "Antonilda", 400.00, 1000.00, 2000.00]
]

df = pd.DataFrame(data, columns=[
    "id", "name", "account_balance", "account_limit", "card_limit"
])

df.to_csv("users.csv", index=False)

df
```

	id	name	account_balance	account_limit	card_limit
0	1	Romilda	700.0	2000.0	4000.0
1	2	Cremilda	512.0	3000.0	4000.0
2	3	Lucinda	784.0	2500.0	3000.0
3	4	Antonilda	400.0	1000.0	2000.0

Próximas etapas: [New interactive sheet](#)

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Próximas etapas: [New interactive sheet](#)

```
users = df.to_dict(orient="records")

users
```

```
[{'id': 1,
  'name': 'Romilda',
  'account_balance': 700.0,
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```
import os
os.environ["OPENAI_API_KEY"] =
```

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```
from openai import OpenAI
import os

client = OpenAI(api_key=os.getenv("OPENAI_API_KEY"))

def generate_message(user):
    prompt = f"""
    Você é um especialista em marketing bancário.

    Cliente: {user['name']}
    Saldo: {user['account_balance']}
    Limite: {user['account_limit']}
    Cartão: {user['card_limit']}

    Crie uma mensagem curta, amigável e personalizada.
    """

    response = client.chat.completions.create(
        model="gpt-4o-mini",
        messages=[{"role": "user", "content": prompt}]
    )

    return response.choices[0].message.content

# Aplicar IA
for user in users:
    user["message"] = generate_message(user)

users
```

```
-----
RateLimitError                                Traceback (most recent call last)
/tmp/ipykernel_7931/3855998753.py in <cell line: 0>()
    25 # Aplicar IA
    26 for user in users:
--> 27     user["message"] = generate_message(user)
    28
    29 users

----- 4 frames -----
/usr/local/lib/python3.12/dist-packages/openai/_base_client.py in request(self, cast_to, options, stream, stream_cls)
   1085
   1086         log.debug("Re-raising status error")
-> 1087         raise self._make_status_error_from_response(err.response) from None
   1088
   1089         break

RateLimitError: Error code: 429 - {'error': {'message': 'You exceeded your current quota, please check your plan and
billing details. For more information on this error, read the docs: https://platform.openai.com/docs/guides/error-codes/api-errors.'}, 'type': 'insufficient_quota', 'param': None, 'code': 'insufficient_quota'}
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Comece a programar ou [gere código](#) com IA.

```
def generate_message(user):
    return f"Olá {user['name']}, temos uma oferta especial de crédito baseada no seu perfil!"
```

```
def generate_message(user):
    saldo = user['account_balance']
    limite = user['account_limit']
    cartao = user['card_limit']

    if saldo < 500 and limite < 2000:
        return f"{user['name']}, libere maior limite!"

    elif limite >= 3000 and cartao >= 4000:
        return f"{user['name']}, aproveite seu perfil premium com vantagens exclusivas!"

    elif saldo > 700:
        return f"{user['name']}, seu bom saldo pode te garantir melhores condições de crédito!"

    else:
        return f"{user['name']}, confira nossas opções de crédito!"
```

```
import json

with open("output.json", "w", encoding="utf-8") as f:
    json.dump(users, f, indent=4, ensure_ascii=False)
```

